From tensor-matrix coefficients to polynomials

Product rule, coefficient extraction, and weighted reduction

Fix $m \geq 1$. Let $V = \mathbb{R}^4$ with basis e_0, e_1, e_2, e_3. Vectors in V are columns; $v = (u, x, y, a)$ is a row covector. For $w \in V^{\otimes m}$, define the homogeneous polynomial

$$P_w(v) := v^{\otimes m} w, \qquad P_{w_1 \otimes \cdots \otimes w_m}(v) = \prod_{j=1}^{m} (v w_j). \tag{1}$$

Thus $P_{e_3^{\otimes m}} = a^m$ and $P_{e_0^{\otimes m}} = u^m$.

Tensor lifts become differential operators. For a matrix $C \in \mathbb{R}^{4\times 4}$, set

$$C^{[m]} = \sum_{s=1}^{m} I^{\otimes(s-1)} \otimes C \otimes I^{\otimes(m-s)}, \qquad D_C = \sum_{j=0}^{3} (vC)_j \partial_{v_j}. \tag{2}$$

The product rule gives, for every pure tensor,

$$D_C \left[\prod_{j=1}^{m} (v w_j) \right] = \sum_{s=1}^{m} (v C w_s) \prod_{j \neq s} (v w_j) = P_{C^{[m]}(w_1 \otimes \cdots \otimes w_m)}(v).$$

By linearity, the exact intertwining identity is therefore

$$\boxed{D_C P_w = P_{C^{[m]} w} \qquad (w \in V^{\otimes m}).} \tag{3}$$

There is no factorial normalization: the encoding uses ordinary pure tensors and ordinary monomials.

The matrices used in the area calculation. Let $E_{ij} e_j = e_i$, with $E_{ij} e_k = 0$ for $k \neq j$, and put

$$C_1 = E_{01} - E_{10} - E_{23} + E_{32}, \qquad C_2 = E_{02} - E_{20} + E_{13} - E_{31}.$$

Since $vC_1 = (-x, u, a, -y)$ and $vC_2 = (-y, -a, u, x)$,

$$\begin{aligned} D_1 &= -x\partial_u + u\partial_x + a\partial_y - y\partial_a, \\ D_2 &= -y\partial_u - a\partial_x + u\partial_y + x\partial_a. \end{aligned} \tag{4}$$

For example, $C_1^{[2]}(e_3 \otimes e_3) = -e_2 \otimes e_3 - e_3 \otimes e_2$, which encodes as $-2ya = D_1(a^2)$. The factor 2 counts the two tensor positions.

A matrix coefficient is an evaluation. Define

$$R_m = (C_1^{[m]})^2 + (C_2^{[m]})^2, \qquad L = D_1^2 + D_2^2.$$

Iterating (3) gives $P_{R_m^m e_3^{\otimes m}} = L^m(a^m)$. Moreover,

$$P_w(1, 0, 0, 0) = (e_0^\top)^{\otimes m} w.$$

Writing $0^m = (0, \ldots, 0)$ and $3^m = (3, \ldots, 3)$, we obtain

$$\boxed{[R_m^m]_{0^m, 3^m} = (L^m a^m)(1, 0, 0, 0).} \tag{5}$$

Weighted reduction to the radial operator

Assign polynomial weights

$$\mathrm{wt}(u) = 0, \qquad \mathrm{wt}(x) = \mathrm{wt}(y) = 1, \qquad \mathrm{wt}(a) = 2.$$

Split each vector field into its weight-lowering and weight-raising parts:

$$\begin{aligned} D_1^- &= u\partial_x - y\partial_a, \quad D_1^+ = -x\partial_u + a\partial_y, \\ D_2^- &= u\partial_y + x\partial_a, \quad D_2^+ = -y\partial_u - a\partial_x. \end{aligned} \tag{6}$$

Each D_j^- sends a weighted-homogeneous polynomial of weight d to one of weight $d-1$, and each D_j^+ sends it to weight $d+1$; zero outputs are allowed.

Expand L^m as a sum of ordered compositions of exactly $2m$ such first-order operators. Starting from a^m, of weight $2m$, a term containing q raising operators has final weight

$$2m + q - (2m - q) = 2q.$$

A weighted-homogeneous polynomial of positive weight vanishes at $(1,0,0,0)$. Consequently, only terms containing lowering operators exclusively can contribute, and

$$(L^m a^m)(1,0,0,0) = \left(((D_1^-)^2 + (D_2^-)^2)^m a^m\right)(1,0,0,0). \tag{7}$$

This argument retains the order of every composition; no operators are assumed to commute. The weights include their polynomial coefficients.

Neither lowering field differentiates u. Thus substitution $u = 1$ intertwines them with $\partial_x - y\partial_a$ and $\partial_y + x\partial_a$. Define

$$\widetilde{H} = (\partial_x - y\partial_a)^2 + (\partial_y + x\partial_a)^2. \tag{8}$$

Equations (5)–(8) imply

$$[R_m^m]_{0^m,3^m} = (\widetilde{H}^m a^m)(0,0,0).$$

For $r = x^2 + y^2$ and a polynomial $p(r,a)$, expansion gives

$$\widetilde{H} = \partial_x^2 + \partial_y^2 + 2(x\partial_y - y\partial_x)\partial_a + (x^2+y^2)\partial_a^2.$$

The angular term annihilates $p(r,a)$, and the Laplacian contributes $4rp_{rr} + 4p_r$. Hence, with $H = 4r\partial_r^2 + 4\partial_r + r\partial_a^2$,

$$\widetilde{H}[p(x^2+y^2,a)] = (Hp)(x^2+y^2,a).$$

Iteration proves the desired deterministic identity:

$$\boxed{[R_m^m]_{0^m,3^m} = (\widetilde{H}^m a^m)(0,0,0) = (H^m a^m)(0,0).} \tag{9}$$

Combining this with the secant contraction identity proved in the companion induction note yields $[R_m^m]_{0^m,3^m} = 2^m m! E_m$, where $E_m = \sec^{(m)}(0)$.

Why symmetrization causes no problem. The map $w \mapsto P_w$ is not injective: for instance, $P_{e_1\otimes e_2 - e_2\otimes e_1} = 0$. However, (3) shows that every $C^{[m]}$ preserves its kernel, and the coefficient functional in (5) factors through the same map. Thus the polynomial representation retains all information needed for this coefficient. The variables commute, but the differential operators need not; indeed, $(D_1D_2 - D_2D_1)a = 2u$.